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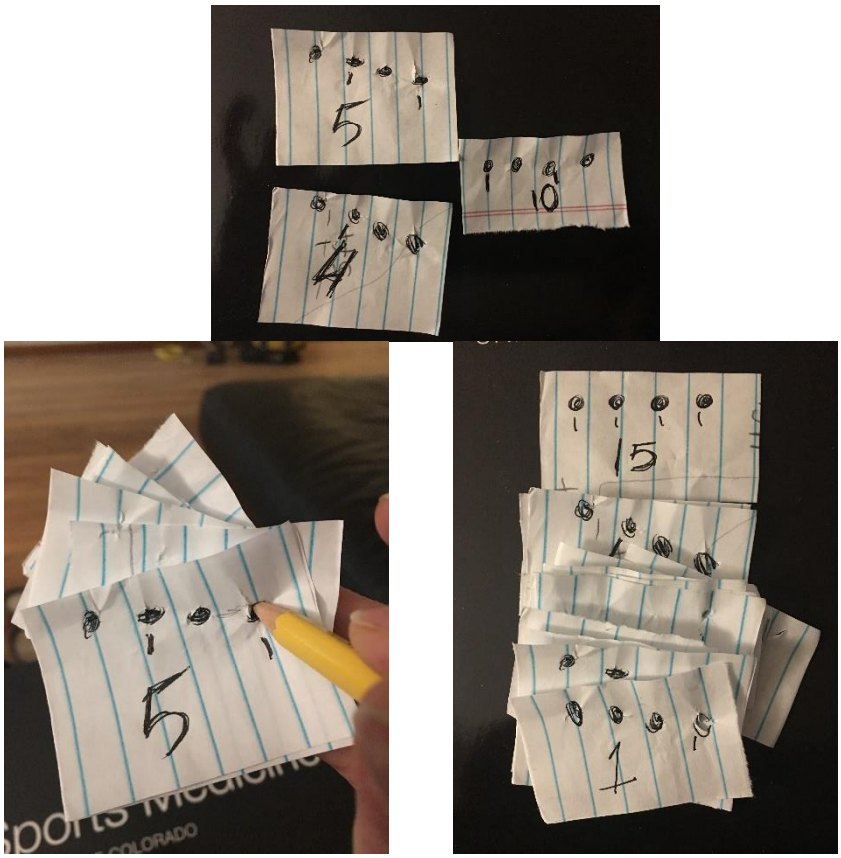
Problem Set 6

1. Old school application of binary numbers



2. Amazing Card Trick

   The amazing card trick works because there are 4 suits and 13 numbers in a 52-card deck. 4 and 13 are both divisors of 52. If the deck is pre-ordered so that the cards alternate red and black suits and in order of suits, then in every case, there is no combination of 4 cards that would contain more or less than two of color and one of every suit. Prediction of the 13th card would not be as successful (i.e. you are relying on more luck than probability) because no previous cards of the family of 4 suits are known therefore you would be able to predict the value of the card but not the suit. For this trick to work, you may leave any single card in any set of 4 face down as long as a total of 13 cards are placed on the table.

3. RSA
   a. What form were keys originally in? Is this some sort of numerical password?
   b. What happens if the key is lost?
   c. What is the capacity of the number of locks Alice can send out? Is there a point in which the number of locks get "lost" or reach capacity or if Alice can no longer receive the locks she has already sent out?

4.2 # 25

$7^{644} \bmod 645$

BINARY EXPANSION OF 644

$$
\begin{aligned}
644 &= 2 \cdot 322 + 0 \\
&= 2 \cdot 2 \cdot 161 + 0 + 0 \\
&= 2^2 \cdot 2 \cdot 80 + 1 + 0 + 0 \\
&= 2^3 \cdot 2 \cdot 40 + 0 + 1 + 0 + 0 \\
&= 2^4 \cdot 2 \cdot 20 + 0 + 0 + 1 + 0 + 0 \\
&= 2^5 \cdot 2 \cdot 10 + 0 + 0 + 0 + 1 + 0 + 0 \\
&= 2^6 \cdot 2 \cdot 5 + 0 + 0 + 0 + 0 + 1 + 0 + 0 \\
&= 2^7 \cdot 2 \cdot 2 + 1 + 0 + 0 + 0 + 0 + 1 + 0 + 0 \\
&= 2^8 \cdot 2 \cdot 1 + 0 + 1 + 0 + 0 + 0 + 0 + 1 + 0 + 0 \\
&= 2^9 \cdot 2 \cdot 0 + \boxed{1010000100}
\end{aligned}
$$

*Initial

x	b	m	power	n
1	7	645	b % m = 7	1010000100

*LOOP

i	n_i	x	power
0	0	1	(7 · 7) % 645 = 49
1	0	1	466
2	1	(1 · 466) mod 645 = 466	436
3	0	466	466
4	0	466	436
5	0	466	466
6	0	466	436
7	1	(466 · 436) mod 645 = 1	466
8	0	1	436
9	1	(1 · 436) mod 645 = $\boxed{436}$	466

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#3 a) $88 = 11 \cdot 2 \cdot 2 \cdot 2$

$\boxed{= 11 \cdot 2^3}$

b) $126 = 2 \times 3 \times 3 \times 7$

$\boxed{= 2 \times 3^2 \times 7}$

c) $729 = 3 \times 3 \times 3 \times 3 \times 3 \times 3$

$\boxed{= 3^6}$

d) $1001 \boxed{= 11 \times 7 \times 13}$

e) $1111 \boxed{= 11 \times 101}$

f) $909{,}090 = 5 \times 2 \times 3 \times 3 \times 3 \times 7 \times 13 \times 37$

$\boxed{= 2 \times 3^3 \times 5 \times 7 \times 13 \times 37}$

#25 a) $3^7 \cdot 5^3 \cdot 7^3 , 2^{11} \cdot 3^5 \cdot 5^9$

$GCD = 2^{\min(0,11)} , 3^{\min(7,5)} , 5^{\min(3,9)} , 7^{\min(3,0)}$

$= 2^0 \cdot 3^5 \cdot 5^3 \cdot 7^0$

$\boxed{= 3^5 \cdot 5^3}$

b) $11 \cdot 13 \cdot 17 , 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$

$\boxed{GCD = 1}$ (no other common divisors exist)

c) $23^{31} , 23^{17}$

$GCD = 23^{\min(31,17)}$

$\boxed{= 23^{17}}$

d) $41 \cdot 43 \cdot 53 , 41 \cdot 43 \cdot 53$

$\boxed{GCD = 41 \cdot 43 \cdot 53}$

e) $3^{13} \cdot 5^{17} , 2^{12} \cdot 7^{21}$

$\boxed{GCD = 1}$ (no other common divisors exist)

f) $1111 , 0$

$\boxed{GCD = 1111}$

4.3 CONT

#27

a) $3^7 \cdot 5^3 \cdot 7^3$, $2^{11} \cdot 3^5 \cdot 5^9$

$LCM = 2^{\max(0,11)} \cdot 3^{\max(7,5)} \cdot 5^{\max(3,9)} \cdot 7^{\max(3,0)}$

$\boxed{= 2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3}$

b) $11 \cdot 13 \cdot 17$, $2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$

$LCM = 2^{\max(0,9)} \cdot 3^{\max(0,7)} \cdot 5^{\max(0,5)} \cdot 7^{\max(0,3)} \cdot 11^{\max(1,0)} \cdot 13^{\max(1,0)} \cdot 17^{\max(1,0)}$

$\boxed{= 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17}$

c) 23^{31}, 23^{17}

$LCM = 23^{\max(31,17)}$

$\boxed{= 23^{31}}$

d) $41 \cdot 43 \cdot 53$, $41 \cdot 43 \cdot 53$

$\boxed{LCM = 41 \cdot 43 \cdot 53}$

e) $3^{13} \cdot 5^{17}$, $2^{12} \cdot 7^{21}$

$LCM = 2^{\max(0,12)} \cdot 3^{\max(13,0)} \cdot 5^{\max(17,0)} \cdot 7^{\max(0,21)}$

$\boxed{= 2^{12} \cdot 3^{13} \cdot 5^{17} \cdot 7^{21}}$

f) 1111, 0

$\boxed{LCM = \text{undefined}}$ 0 has no multiplier

4.3 CONT

#33

a) GCD(12, 18)

$= GCD(12 \bmod 18, 18)$

$12 \bmod 18 = 12$

$= GCD(18, 12)$

$= GCD(\underbrace{18 \bmod 12}, 12)$

$= GCD(12, 6)$

$= GCD(\underbrace{12 \bmod 6}, 6)$

$= GCD(6, 0)$

$= \boxed{6}$

b) GCD(111, 201)

$= GCD(\underbrace{111 \bmod 201}, 201)$

$= GCD(201, 111)$

$= GCD(201 \bmod 111, 111)$

$= GCD(111, 90)$

$= GCD(111 \bmod 90, 90)$

$= GCD(90, 21)$

$= GCD(90 \bmod 21, 21)$

$= GCD(21, 6)$

$= GCD(21 \bmod 6, 6)$

$= GCD(6, 3)$

$= GCD(6 \bmod 3, 3)$

$= GCD(3, 0) = \boxed{3}$

4.3 CONT

33 CONT

c) GCD(1001, 1331)

= GCD(1001 mod 1331, 1331)

= GCD(1331, 1001)

= GCD(1331 mod 1001, 1001)

= GCD(1001, 330)

= GCD(1001 mod 330, 330)

= GCD(330, 11)

= GCD(330 mod 11, 11)

= GCD(11, 0)

= $\boxed{11}$

d) GCD(12345, 54321)

= GCD(12345 mod 54321, 54321)

= GCD(54321, 12345)

= GCD(54321 mod 12345, 12345)

= GCD(12345, 4941)

= GCD(12345 mod 4941, 4941)

= GCD(4941, 2463)

= GCD(4941 mod 2463, 2463)

= GCD(2463, 15)

= GCD(2463 mod 15, 15)

= GCD(15, 3) = GCD(15 mod 3, 3)

= GCD(3, 0) = $\boxed{3}$

4.3 CONT

#33 CONT

e) GCD(1000, 5040)

= GCD(1000 mod 5040, 5040)

= GCD(5040, 1000)

= GCD(5040 mod 1000, 1000)

= GCD(1000, 40)

= GCD(1000 mod 40, 40)

= GCD(40, 0)

= $\boxed{40}$

f) GCD(9888, 6060)

= GCD(9888 mod 6060, 6060)

= GCD(6060, 3828)

= GCD(6060 mod 3828, 3828)

= GCD(3828, 2232)

= GCD(3828 mod 2232, 2232)

= GCD(2232, 1596)

= GCD(2232 mod 1596, 1596)

= GCD(1596, 636)

= GCD(1596 mod 636, 636)

= GCD(636, 324)

= GCD(636 mod 324, 324)

= GCD(324, 312) = GCD(324 mod 312, 312)

= GCD(312, 12) = GCD(312 mod 12, 12)

= GCD(12, 0) = $\boxed{12}$

7. Sieve of Eratosthenes – theorem 2
    a. Prove that 100 must have a prime divisor <= 10

        Let n = 100

        Because n is a composite number it must also be true that n = a*b for integers a and b such that 1 < a < n and 1 < b < n.

        Let $\sqrt{a} * \sqrt{b} = \sqrt{n}$ because a and b are divisors of n

        $\sqrt{a} * \sqrt{b} = \sqrt{100}$

        $\sqrt{a} * \sqrt{b} = 10$

        Therefore, either a or b must be < 10 and either a or b (the divisors) are prime.

    b. This could be used in cryptography as when two primes are multiplied it would be very difficult to factor these numbers into their primes. One would have to test all primes less than the number and see if a multiplication resulted in the desired number.